Report: Fraudulent Claim Detection

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1. Problem Statement:

Global Insure, a leading insurance company, processes thousands of claims annually. However, a significant percentage of these claims turn out to be fraudulent, resulting in considerable financial losses. The company's current process for identifying fraudulent claims involves manual inspections, which is time-consuming and inefficient. Fraudulent claims are often detected too late in the process, after the company has already paid out significant amounts. Global Insure wants to improve its fraud detection process using data-driven insights to classify claims as fraudulent or legitimate early in the approval process. This would minimise financial losses and optimise the overall claims handling process.

2. Methodology:

Our approach to developing a predictive model for identifying fraudulent insurance claims followed a systematic, data-driven process aimed at improving efficiency and reducing losses.

Key stages of the methodology included:

I. Objective:

- Clearly understanding the business challenge: the cost of fraud and the limitations of manual detection.
- Setting the objective: building a model to classify claims as fraudulent or legitimate early in the process.

II. Data Preparation:

- *Data Cleaning:*
  - Identifying and handling missing data (e.g., filling or removing).
  - Removing irrelevant or illogical data points and columns (e.g., identifiers, entries with impossible values).

- ***Exploratory Data Analysis (EDA):***
  - Analyzing individual data points to understand their characteristics and distributions.

- Investigating relationships between data points and fraud to find patterns.
- Identifying the significant imbalance between legitimate and fraudulent claims.

III. Data Splitting and Balancing:

- Dividing the data into training (70%) and validation (30%) sets to simulate real-world performance.
- Ensuring both sets have a representative mix of claim types (stratification).
- Using SMOTE (Synthetic Minority Over-sampling Technique) to balance the training data by creating synthetic examples of fraudulent claims, preventing model bias. The validation data was kept in its original, imbalanced state.

IV. Feature Engineering:

- Creating new, potentially more informative data points from existing ones (e.g., extracting month from date, calculating claim ratios, creating flags for high claim amounts).
- This aimed to give the models better information to work with.

V. Feature Transformation:

- Converting categorical information (like incident type) into a numerical format using dummy variables for model compatibility. Infrequent categories were grouped first.
- Scaling numerical data to a common range so no single feature dominates the model based purely on its value size. Consistent scaling was applied to both training and validation data.

VI. **Model Building and Selection:**

- Selecting two candidate models: Logistic Regression and Random Forest.
- Using data-driven techniques (RFECV for Logistic Regression, Feature Importance for Random Forest) to select the most predictive subset of features for each model type.

VII. Model Optimization (Random Forest):

- Employing Grid Search with cross-validation to systematically find the best settings (hyperparameters) for the Random Forest model, optimizing for the F1 score which balances identifying fraud against minimizing false alarms.

VIII. Prediction and Evaluation:

- Using the trained models (tuned Random Forest and Logistic Regression with an optimal cutoff) to predict the probability of fraud on the unseen validation data.
- Evaluating model performance using key metrics relevant to fraud detection: Accuracy (overall correct), Balanced Accuracy (fairness across classes), Confusion Matrix (breakdown of correct/incorrect predictions), Sensitivity/Recall (finding actual fraud), Specificity (correctly identifying non-fraud), Precision (accuracy of fraud predictions), and F1 Score (balance of finding fraud and being accurate).
- Comparing the performance of the two models on the validation data to determine the most effective approach for deployment.

This comprehensive process ensures the resulting model is built on clean, relevant data, is trained effectively to handle the challenge of rare fraudulent cases, and is evaluated realistically on unseen data to provide confidence in its real-world applicability.

3. Analytical Techniques Used

Our approach to building a predictive model for detecting fraudulent insurance claims involved applying various data science and machine learning techniques. Each technique served a specific purpose in preparing the data, building the models, and evaluating their effectiveness.

I. Data Cleaning and Preparation:

- Handling Missing Data (Imputation):
  - Filling in gaps in the data, often by using the most common value for categories, to ensure complete records for analysis.
- Removing Redundant Information:
  - Identifying and removing columns that were unique identifiers or contained illogical values, as they do not help the model find general patterns.

II. Understanding the Data (Exploratory Data Analysis - EDA):

- Visualizations & Statistics:
  - Using charts (like histograms, bar charts, box plots) and numerical summaries (like mean, median, percentages) to understand data distributions and key characteristics of features.
- Relationship Analysis:
  - Examining how individual features relate to whether a claim is fraudulent, using visuals and statistical tests to identify potential predictors.
- Checking for Imbalance:
  - Analyzing the target variable to confirm if one outcome (e.g., legitimate claims) is much more common than the other (fraudulent claims).

III. Addressing Data Challenges:

- Handling Imbalance (SMOTE):
  - Applying a technique (SMOTE) to the training data to create synthetic examples of the less frequent outcome (fraud) so the model has enough examples to learn from, preventing it from ignoring fraud cases. Note: Applied only to training data.

IV. Feature Engineering & Transformation:

- Creating New Features:
  - Generating new, potentially more informative features from existing data (e.g., extracting date components, calculating ratios like claim amount per vehicle).
- Making Data Model-Ready (Encoding & Scaling):
  - Converting categorical data into a numerical format (one-hot encoding) and adjusting the scale of numerical data (standardization) so models can process them effectively.

V. Building Predictive Models:

- Logistic Regression:
    - Using a statistical model that estimates the probability of fraud based on features, known for its interpretability.
- Random Forest:
    - Employing a powerful model that combines multiple decision trees to make predictions, known for its accuracy and ability to capture complex relationships.

VI. Refining the Models:

- Feature Selection:
    - Identifying the most relevant subset of features for each model type (using RFECV for Logistic Regression and Feature Importance for Random Forest) to improve performance and focus the models.
- Hyperparameter Tuning (Grid Search):
    - Systematically testing different internal settings for the Random Forest model to find the combination that gives the best performance (specifically optimizing for the F1 score) on unseen data during the tuning process.

VII. Evaluating Model Performance:

- Prediction on Unseen Data:
    - Applying the trained models to a separate set of data they haven't seen before (validation data) to get a realistic measure of their performance.
- Performance Metrics:
    - Using a set of standard measures (Accuracy, Balanced Accuracy, Confusion Matrix, Sensitivity, Specificity, Precision, F1 Score) to assess how well each model identifies fraudulent and legitimate claims. These metrics provide a detailed understanding of where the model succeeds and where it makes errors.
- Optimal Cutoff Analysis:
    - For models that output probabilities (like Logistic Regression or Random Forest), finding the best threshold to classify claims as fraud vs. non-fraud, balancing the goal of catching fraud against the risk of false alarms based on business needs.

By applying these techniques, we developed and evaluated models designed to effectively predict fraudulent insurance claims, providing a data-driven tool for Global Insure.

4. <u>Data Visualization</u>

Data visualization played a crucial role throughout this project. It allowed us to see and understand complex patterns, distributions, and relationships within the data that would be difficult or impossible to grasp from raw numbers alone. By creating visual representations, we gained insights that guided our data cleaning, feature engineering, model building, and evaluation processes.

Here are the key types of data visualizations we used and why they were important:

I. Histograms:

- What they show: The distribution of numerical data. They divide the data into bins and show how many data points fall into each bin, revealing the shape of the data (e.g., whether it's spread out, clustered in the middle, or skewed to one side).
- Why we used them: To understand the typical range, variability, and frequency of values for numerical features like age, claim amounts, or incident hour. This helped us see if data was concentrated in certain ranges or if there were unusual peaks or gaps. When overlaid for fraud vs. non-fraud cases, they helped us visually compare distributions between the two groups.

II. Box Plots:

- What they show: A summary of the distribution of numerical data, highlighting the median (middle value), quartiles (dividing the data into quarters), and potential outliers.
- Why we used them: To quickly identify the central tendency, spread, and presence of extreme values (outliers) in numerical features. Comparing box plots side-by-side for fraud vs. non-fraud helped us see if the typical values or the spread of a numerical feature differed significantly between the two groups.

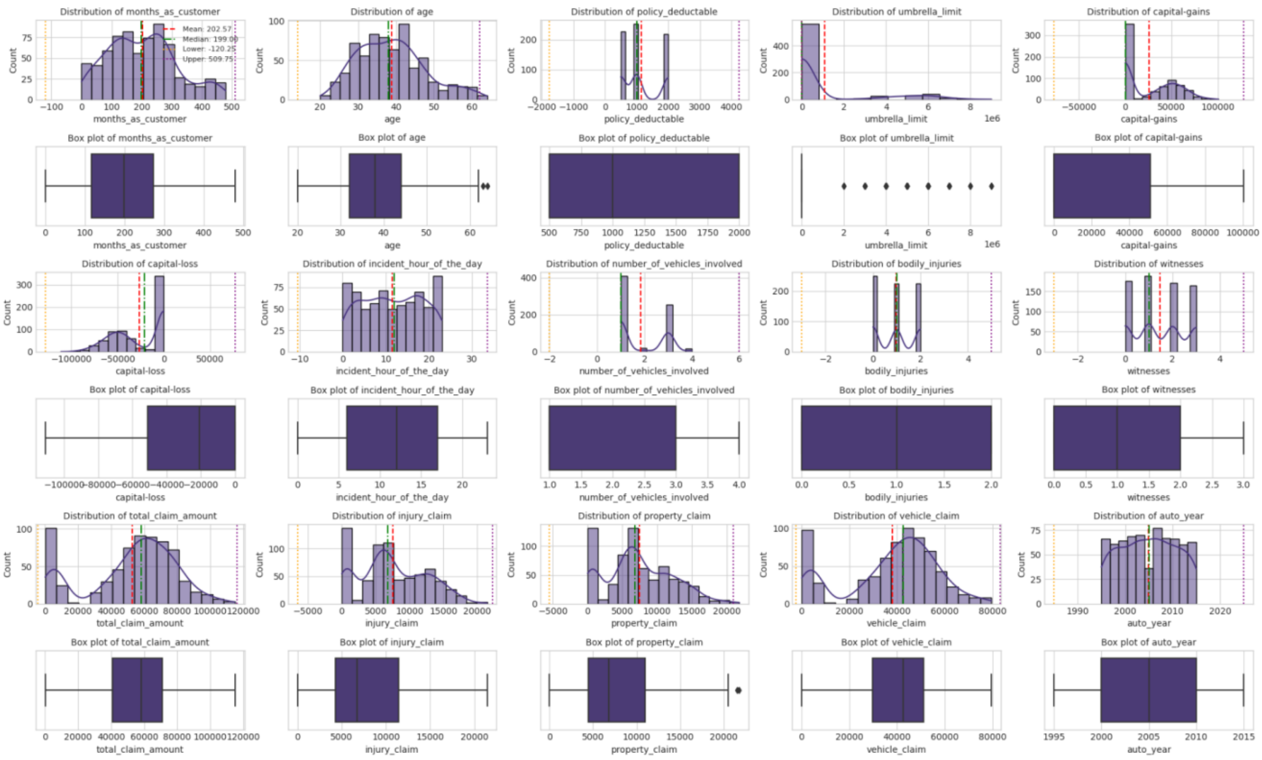


Fig. 1. Univariate Analysis (4.1.2)

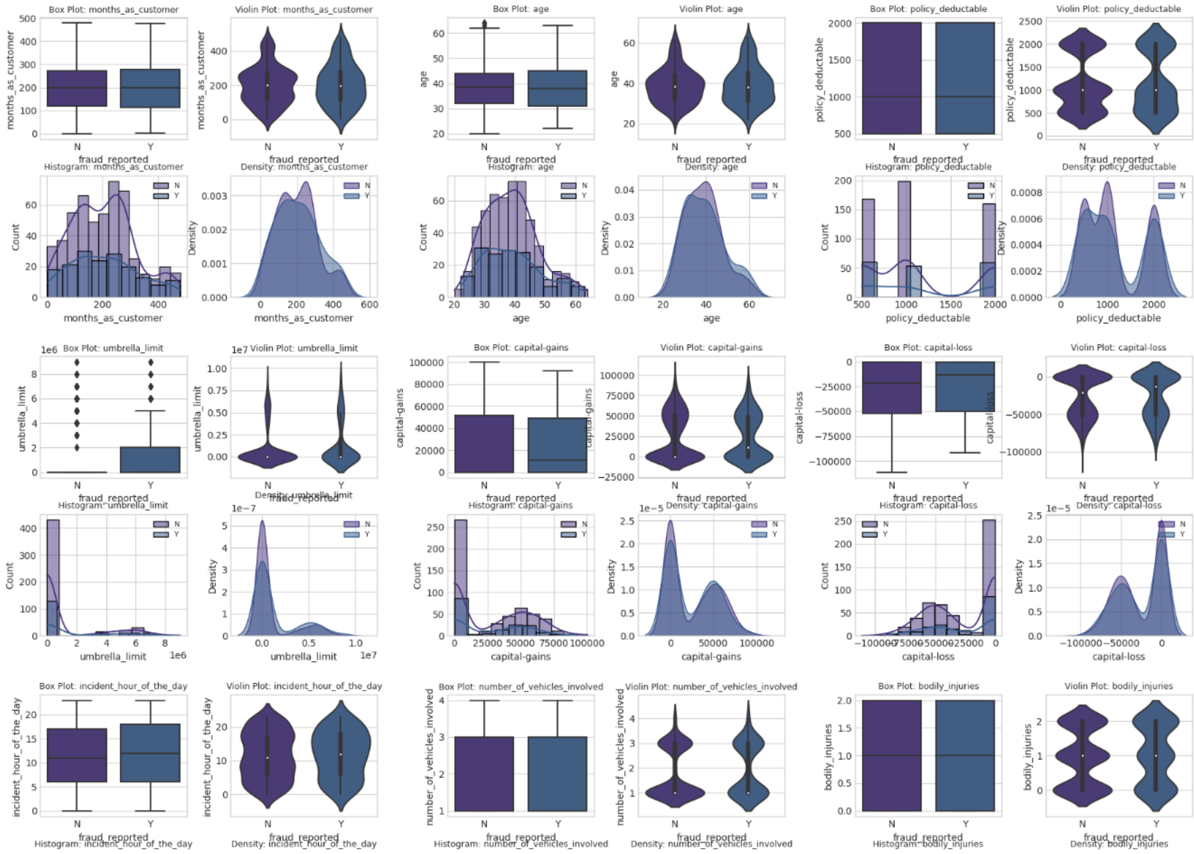


Fig. 2. Bivariate Analysis (4.4.2)

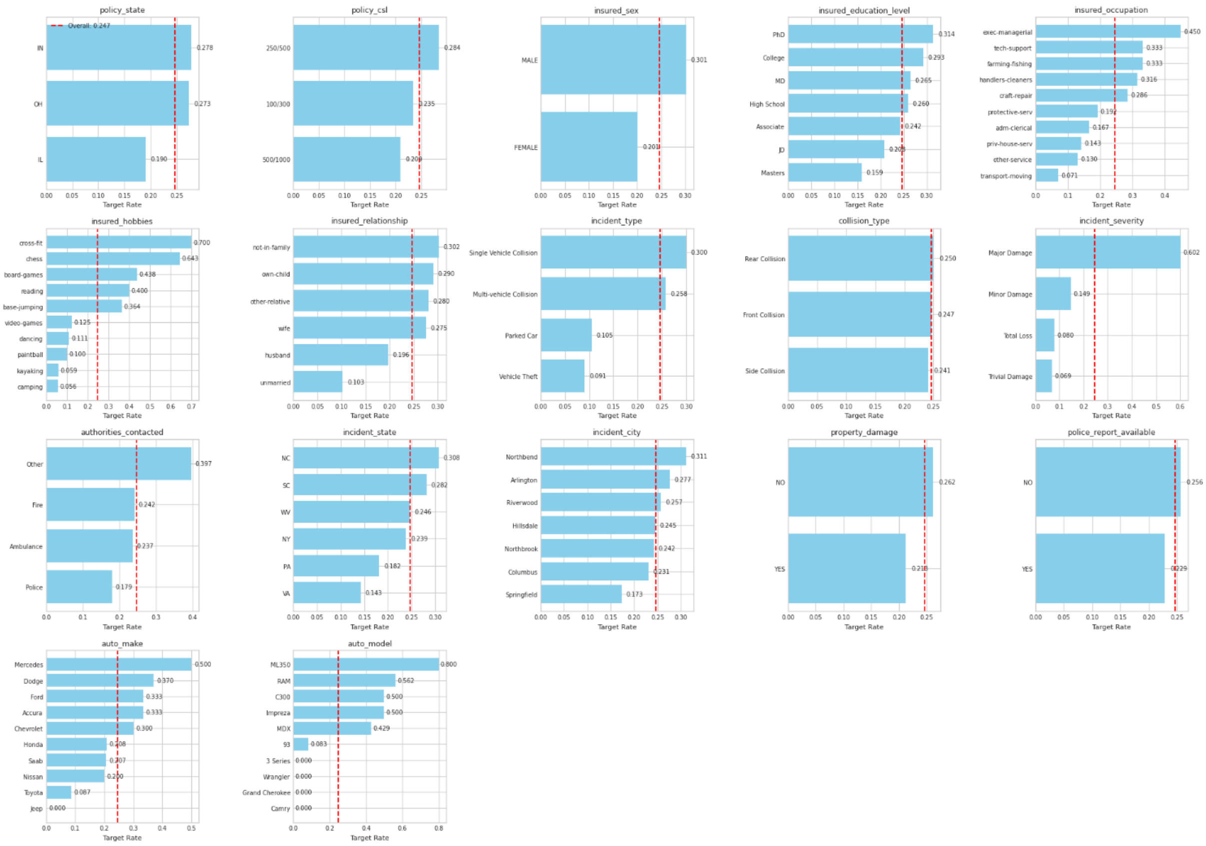


Fig. 3. Target likelihood analysis for categorical variables

III. Bar Charts:

- What they show: The counts or proportions of different categories within a categorical feature.
- Why we used them: To see the frequency of different policy states, incident types, occupations, etc. They were particularly useful for visualizing the class balance (or imbalance) in the target variable (fraud vs. non-fraud) and for showing the fraud likelihood across different categories during bivariate analysis.

IV. Density Plots (KDE - Kernel Density Estimate):

- What they show: A smoothed version of a histogram, providing a continuous view of the distribution of numerical data.
- Why we used them: To get a clearer picture of the shape of numerical distributions, especially when comparing the distributions of a feature between fraudulent and legitimate claims. They are good at showing overlaps or separations between groups.

V. ROC Curves (Receiver Operating Characteristic Curves):

- What they show: The performance of a classification model across all possible classification thresholds. It plots the True Positive Rate (Sensitivity - how well it finds fraud) against the False Positive Rate (1 - Specificity - how often it flags non-fraud as fraud).
- Why we used them: To understand the trade-off between correctly identifying fraud and incorrectly flagging legitimate claims. The Area Under the Curve (AUC) value provides a single measure of the model's overall ability to distinguish between the two classes.

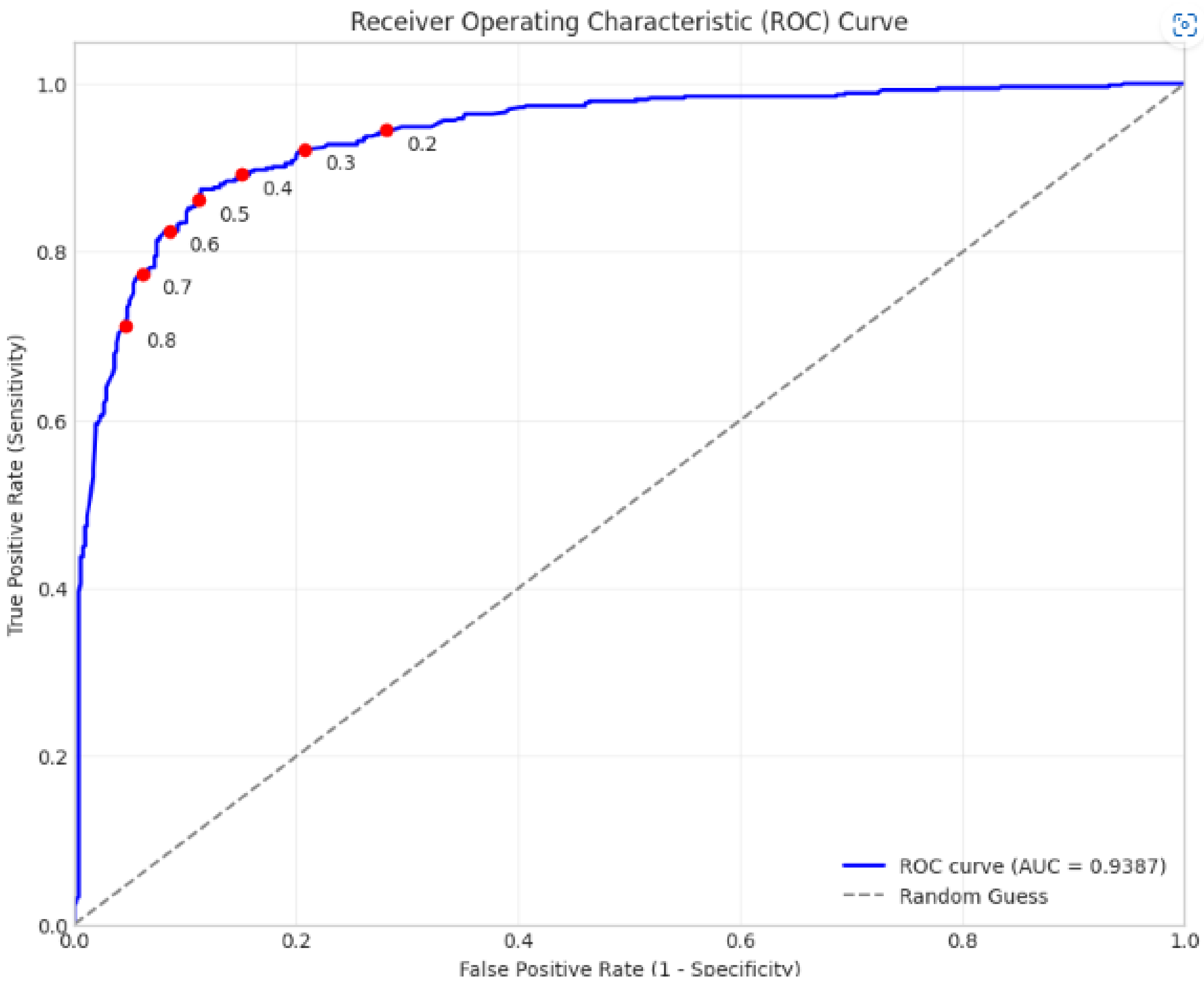


Fig. 4. ROC Curve (7.3.1)

VI. Precision-Recall Curves:

- What they show: The trade-off between Precision (how accurate the fraud predictions are) and Recall (Sensitivity - how many actual fraud cases are found) at different classification thresholds.
- Why we used them: These curves are especially informative for imbalanced datasets. They highlight how Precision drops as we try to increase Recall (catch more fraud). The Average Precision (AP) score summarizes performance focused specifically on the positive class (fraud).

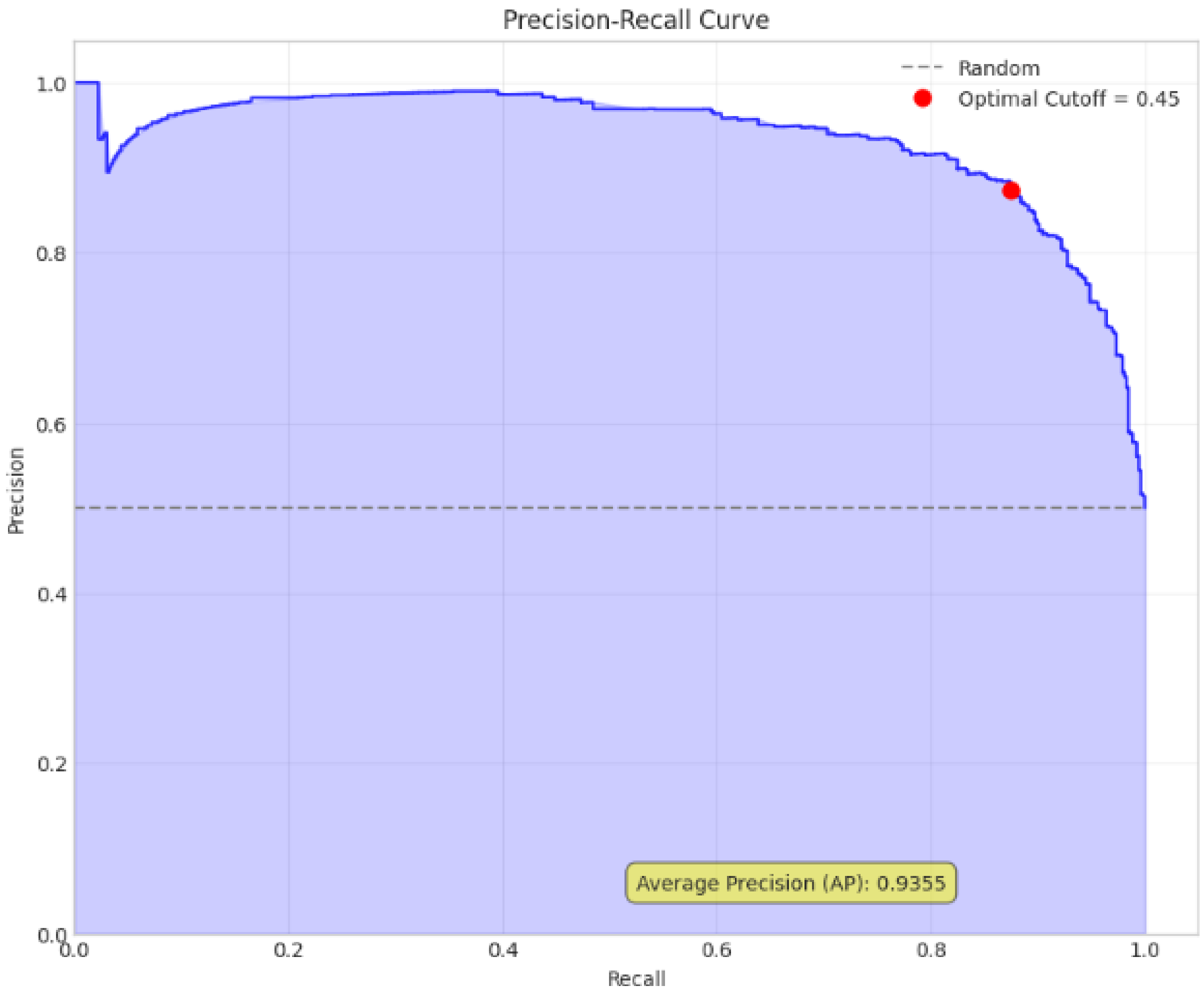


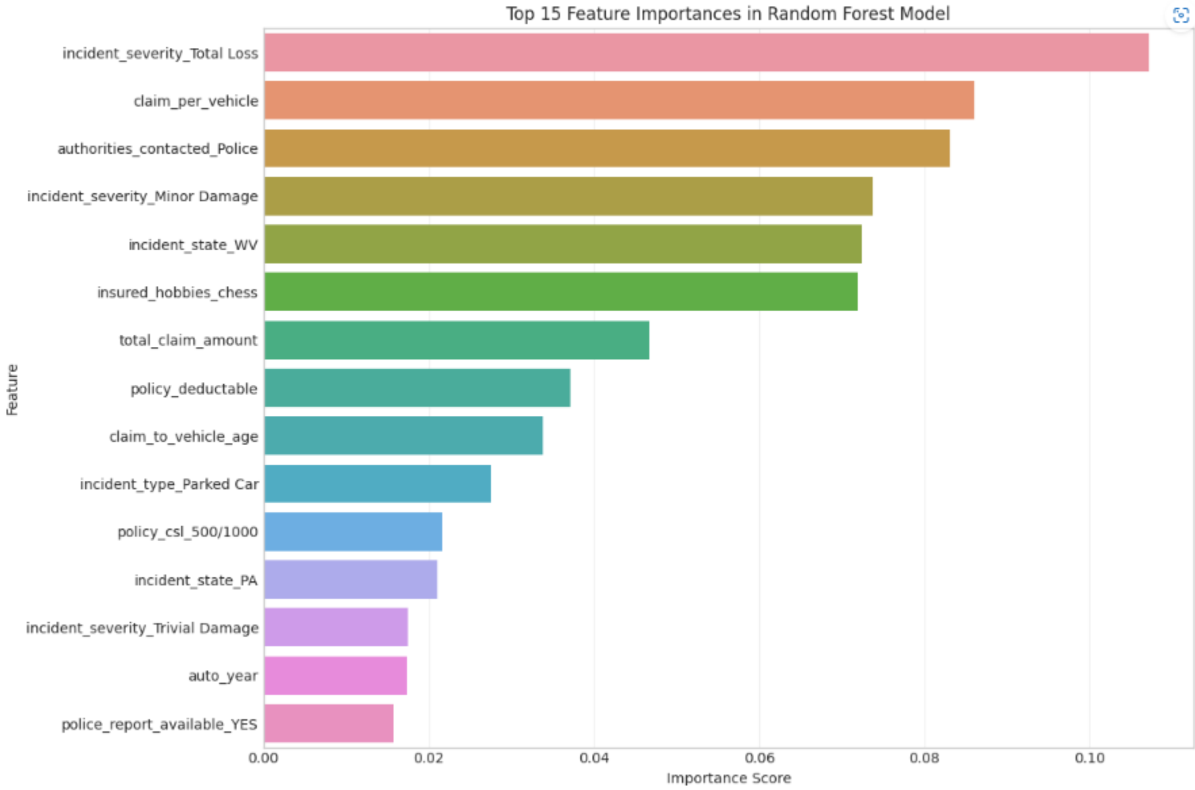
Fig. 5. Precision Recall Curve (7.3.9)

VII. Confusion Matrices (Heatmaps):

- What they show: A table summarizing the model's predictions versus the actual outcomes, broken down into True Positives, True Negatives, False Positives, and False Negatives. Visualized as a heatmap, it shows these counts clearly.
- Why we used them: To get a direct and intuitive view of exactly where the model is making errors (false positives and false negatives). This is crucial for understanding the practical implications of the model's performance.

VIII. Feature Importance Plots:

- What they show: For models like Random Forest, these plots rank features based on how much they contributed to the model's predictive power.
- Why we used them: To identify the most influential factors that the model used to predict fraud. This provides valuable insights into what drives fraudulent behavior according to the data and the model.



By incorporating these visualizations throughout the project, we were able to gain deep insights into the data, make informed decisions about feature engineering and model selection, and clearly communicate the models' performance and limitations.

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5. Key Insights

Through the process of analyzing the data, building models, and evaluating their performance, we gained several key insights into the nature of fraudulent insurance claims within the provided dataset and developed a model capable of assisting Global Insure in detecting them.

I. Understanding the Nature of Fraud in the Data:

- Fraudulent claims represent a significant minority (approximately 25%) of the historical claims, highlighting the real-world challenge of identifying these rarer events.
- Certain characteristics are strongly associated with fraudulent claims:
  - *Incident Severity*: Claims involving "Major Damage," "Total Loss," or even "Trivial Damage" were found to have significantly different likelihoods of fraud compared to "Minor Damage." Specifically, "Major Damage" was strongly linked to higher fraud rates, while "Trivial Damage" and "Total Loss" were associated with lower rates (relative to the baseline/other categories).

- *Insured Hobbies*: Surprisingly, certain hobbies showed a strong statistical association with fraud likelihood, particularly "chess" and "cross-fit" being associated with higher fraud rates. This highlights potentially unexpected patterns in the data.
- *Claim Amounts*: The total claim amount and its components (vehicle, property, injury claims) showed statistically significant differences in their distributions between fraudulent and legitimate claims. Fraudulent claims, on average, tended to have higher total claim amounts.
- *Incident Details*: The type of incident (e.g., "Single Vehicle Collision," "Vehicle Theft") and the state where the incident occurred also showed moderate associations with fraud likelihood.

II. Identifying the Most Predictive Factors:

- Feature importance analysis from the Random Forest model, and the selected features from Logistic Regression (RFECV), largely aligned with the bivariate analysis findings.
- Key predictors identified include aspects of:
  - Incident Severity and Type
  - Claim Amounts and Engineered features derived from them (like claim per vehicle)
  - Specific Insured Hobbies
  - Authorities Contacted (e.g., Police)
  - Geographic Location of the Incident (State)
  - Date/Time of Incident (Month)
  - Vehicle Details (Make)
- These features are the most influential indicators for predicting fraud based on the models developed.

III. Model Performance and Capability:

- The tuned Random Forest model performed well on the unseen validation data, demonstrating its ability to generalize beyond the training examples.
- Key performance highlights on the validation data (using a standard 0.5 classification cutoff) include:
  - *Overall Accuracy of 80.33%:* The model correctly classified over 80% of claims.
  - *Balanced Accuracy of 73.77%:* This metric confirms that the model performs reasonably well across both legitimate and fraudulent classes, not just by predicting the majority class.
  - *Sensitivity (Recall) of 60.81%:* The model successfully identified over 60% of the actual fraudulent claims in the validation set. This means a significant portion of fraudulent activity can be flagged.
  - *Specificity of 86.73%:* The model correctly identified over 86% of legitimate claims as legitimate, helping to minimize unnecessary investigations into valid claims.
  - *Precision of 60.00%:* When the model flags a claim as potentially fraudulent, it is correct 60% of the time. This provides a good signal-to-noise ratio for investigators.
  - *F1 Score of 60.40%:*

IV. Implications for Global Insure:

- Proactive Fraud Detection: The developed model provides a data-driven tool to assess the likelihood of fraud for incoming claims automatically.
- Improved Efficiency: By flagging claims with a high probability of fraud, Global Insure can focus its investigative resources more effectively on the most suspicious cases, reducing the need for manual review of every claim.
- Reduced Financial Losses: Identifying fraudulent claims earlier can prevent significant payouts, directly impacting the company's bottom line.
- Actionable Intelligence: The key predictive features identified by the model offer insights into characteristics associated with fraud, which can potentially inform risk assessment, policy underwriting rules, and operational procedures to mitigate fraud risk proactively.

In conclusion, this project successfully built and validated a predictive model that offers a significant opportunity for Global Insure to enhance its fraud detection capabilities, improve operational efficiency, and reduce financial exposure to fraudulent claims. The Random Forest model, leveraging insights from historical data and engineered features, provides a robust tool for identifying suspicious claims.

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6. <u>Recommendations and Business Implications</u>

Based on our analysis and model performance, we recommend implementing a data-driven approach to fraud detection. This offers significant benefits for Global Insure's efficiency and financial health.

Key Recommendations and Implications:

I. Deploy the Random Forest Model

- What to do: Integrate the model to score new claims for fraud probability.
- Implication: Prioritize high-score claims for investigation, streamlining the process and focusing resources.

II. Use a Flexible Fraud Threshold

- What to do: Define the cutoff for flagging claims based on whether catching more fraud or reducing false alarms is the priority.
- Implication: Tailor the detection process to align with current business goals (e.g., maximize fraud catch or minimize investigation costs).

III. Highlight Key Fraud Indicators

- What to do: When a claim is flagged, show investigators the specific features (like incident severity, claim amounts, certain hobbies) that contributed to the high score.
- Implication: Provide investigators with actionable insights, speeding up reviews and improving decision-making.

IV. Inform Risk Assessment & Underwriting

- What to do: Share findings on high-risk features with teams responsible for assessing risk and setting policy terms.
- Implication: Potentially refine underwriting rules or pricing to mitigate fraud risk proactively before claims occur.

V. Continuously Monitor and Update the Model

- What to do: Regularly check model performance and retrain it with new data as fraud patterns change.
- Implication: Ensure the model remains accurate and effective over time, maintaining its value as a detection tool.

VI. Consider Explanations for Model Flags

- What to do: Explore ways to provide simple reasons why a claim was flagged (e.g., "Claim amount is unusually high for this vehicle type").
- Implication: Build trust and confidence among claims staff using the AI tool.

VII. Track Business Impact

- What to do: Measure how the model affects metrics like flagged claims, confirmed fraud cases, financial savings, and processing times.
- Implication: Quantify the return on investment and demonstrate the value of the AI solution.

Implementing these steps can significantly enhance Global Insure's ability to combat fraud, leading to improved efficiency and reduced losses.

7. Assumptions Made:

Developing this predictive model involved certain assumptions about the data and future claims. Understanding these is key to interpreting the model's results and planning for its use.

Key Assumptions:

I. Assumption 1: Future Claims will resemble Historical Data.

- Meaning: Patterns and relationships seen in past claims are expected to continue.

- Implication: Significant changes in fraud tactics or claim types could reduce model accuracy over time.

II. Assumption 2: Cleaned Data is Sufficiently Reliable.

- Meaning: Data cleaning steps (like filling missing values) didn't introduce significant misleading information.
- Implication: Uncorrected errors or biases in the original data could impact model learning and predictions.

III. Assumption 3: Available Data Contains Key Predictors.

- Meaning: The dataset includes the most important factors for identifying fraud.
- Implication: If crucial fraud indicators are missing from the data, the model's ability to detect fraud will be limited.

IV. Assumption 4: Historical Fraud Labels are Accurate.

- Meaning: Past claims were correctly marked as fraudulent or legitimate.
- Implication: Errors in historical labeling will cause the model to learn incorrectly, affecting its accuracy on new claims.

V. Assumption 5: The Relationships between Features and Fraud are Stable.

- Meaning: How features like incident type or claim amount relate to fraud doesn't change rapidly.
- Implication: Quick shifts in claim characteristics or fraud methods could make the model's learned relationships outdated.

VI. Assumption 6: Chosen Modeling Techniques are Suitable.

- Meaning: Logistic Regression and Random Forest are appropriate for finding fraud patterns in this data.
- Implication: If the underlying fraud patterns are very complex in ways these models can't capture, other techniques might perform better.

Being aware of these assumptions helps Global Insure understand the model's context and the need for ongoing monitoring and potential updates.